

Gareth Beall

Lead AI/ML Engineer

Clinical pharmacist turned AI engineer, building retrieval systems that can survive real governance, safety, and operational constraints.

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PROFILE

Lead AI/ML engineer and clinical pharmacist with fifteen years of clinical practice, including intensive care. I build production AI systems for settings where evidence, safety, governance, and operational reliability matter.

EXPERIENCE

Lead AI/ML Engineer / Sunshine Coast Hospital and Health Service

February 2025 to present | Hybrid, Sunshine Coast, Queensland

- Conceived and built the initial IRIS clinical RAG system, then led its engineering into a governed Queensland Health delivery context for clinical evaluation and user acceptance testing.
- Built auditable ingestion for PDF, Word, and web content, including text, tables, images, source identity, version detection, and quarantine paths for material that should not be retrieved.
- Built hybrid lexical and semantic retrieval, reranking, row-level table retrieval, citation continuity, and evaluation against realistic clinical questions.
- Operate self-hosted model-serving and document-processing workloads on GPU infrastructure, covering containers, model loading, concurrency, health, latency, and throughput.
- Treat prompts, retrieval, reranking, and model changes as experiments, using clinician-graded and automated evaluations to study failures and decide what to change next.
- Built the surrounding delivery system across APIs, workers, queues, PostgreSQL, object storage, administration tools, audit records, observability, migrations, backup, restore, and controlled releases.

Founder and Software Engineer / E2 Apps

September 2022 to present | Remote, Sunshine Coast, Queensland

- Built and operate Opioid Conversion Calculator, a focused medication-safety product for health professionals, including its clinical content, Flutter app, backend, analytics, subscriptions, releases, monitoring, and support.
- Shipped Namely, a preference-led baby-name discovery product, and My Restored Photo, an AI-assisted photo-restoration service with uploads, processing, editing, payments, private storage, and scheduled deletion.

Founder and AI Engineer / Inky Health

August 2026 to present | Remote, Sunshine Coast, Queensland

- Develop private, self-hosted document AI for health and government settings where sensitive data cannot leave the organisation's network, including document search, structured extraction, redaction, and agent workflows.

SELECTED SYSTEM

IRIS / Clinical retrieval-augmented generation

A governed clinical RAG system designed for the realities of healthcare: traceable sources, controlled ingestion, hybrid retrieval, reranking, evaluation, self-hosted inference, safety controls, and disciplined release operations.

INGESTION	Reviewed sources, auditable processing, structured metadata, and quarantine paths for material that should not be retrieved.
RETRIEVAL	Hybrid lexical and semantic search, reranking, citation continuity, and evaluation against realistic clinical questions.
INFERENCE	Self-hosted model serving on GPU infrastructure, with performance, privacy, availability, and operational constraints treated as engineering inputs.
SAFETY	Guardrails, scope controls, evidence display, evaluation suites, and respectful handling of uncertainty rather than a veneer of confidence.
DELIVERY	Security, UAT, release gates, observability, incident response, documentation, and the work needed to keep an AI system useful after launch.

CAPABILITIES

Clinical AI	RAG, agent workflows, prompt and tool design, model evaluation, safety controls, and fine-tuning.
Retrieval and data	Hybrid lexical and semantic search, embeddings, vector databases, reranking, PostgreSQL, and object storage.
Software and operations	Python, TypeScript, APIs, workers, queues, vLLM, Linux, Docker, Podman, CI/CD, monitoring, backup, and release management.

SELECTED PRODUCT WORK

Opioid Conversion Calculator <i>Clinical product</i> A practical conversion tool for health professionals, translating specialist medication knowledge into a focused, usable interface.	Namely <i>Consumer product</i> A baby-name discovery product shaped around playful exploration, preference signals, and the small details that make decision tools feel human.	My Restored Photo <i>AI product</i> A photo-restoration experience that applies modern image technology to something emotionally legible: recovering damaged personal photographs.
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EDUCATION

Bachelor of Pharmacy The University of Queensland	2008
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BRING ME THE PROBLEM THAT NEEDS BOTH JUDGMENT AND ENGINEERING.